

Advertising Sales Prediction

Multiple Linear Regression Analysis

A practical analysis of how TV, Radio, and Newspaper advertising spending is related to product sales, followed by a sales prediction for a planned advertising budget.

200

DATASET ROWS

4

COLUMNS

80 / 20

TRAIN / TEST SPLIT

0.906TEST R²

1. EXECUTIVE SUMMARY

The project uses Multiple Linear Regression to estimate Sales from three advertising channels: TV, Radio, and Newspaper. The model was trained on 80% of the 200 records and evaluated on the remaining 20%. The test R² is 0.906, showing that the model explains about 90.6% of the variation in unseen sales values.

For the planned budget of TV = 150, Radio = 20, and Newspaper = 30, the model predicts approximately **15.04 sales units**. Among the raw regression coefficients, Radio has the strongest impact and Newspaper has the least.

2. DATASET AND METHOD

Item	Details
Dataset	Advertising dataset
Records	200
Input features	TV, Radio, Newspaper
Target	Sales
Missing values	None
Model	Multiple Linear Regression
Data split	80% training / 20% testing
Random state	42

The dataset was first loaded and checked for its shape, columns, data types, summary statistics, and missing values. The three advertising columns were then selected as model inputs and Sales was used as the target.

Model Building & Evaluation

3. MODEL SPECIFICATION

The trained model learns a linear relationship between advertising spending and sales. The fitted equation from the submitted notebook is:

Sales = 4.7141 + 0.0545 × TV + 0.1009 × Radio + 0.0043 × Newspaper

4. MODEL COEFFICIENTS

Term	Coefficient	Meaning
Intercept	4.7141	Baseline predicted sales when inputs are zero
TV	0.0545	Positive contribution to predicted sales
Radio	0.1009	Largest positive coefficient
Newspaper	0.0043	Smallest positive coefficient

5. MODEL EVALUATION

Metric	Value	Simple interpretation
MAE	1.2748	Average absolute prediction error
MSE	2.9078	Average squared prediction error
RMSE	1.7052	Typical prediction error size
R²	0.9059	About 90.6% of sales variation is explained

1.27

MAE

1.71

RMSE

0.906

R² SCORE

0.91

APPROX. MODEL FIT

The evaluation results show that the model performs reasonably well on unseen data. The R² score is close to 0.91, while the RMSE indicates an average prediction error of about 1.71 sales units.

Prediction & Visual Analysis

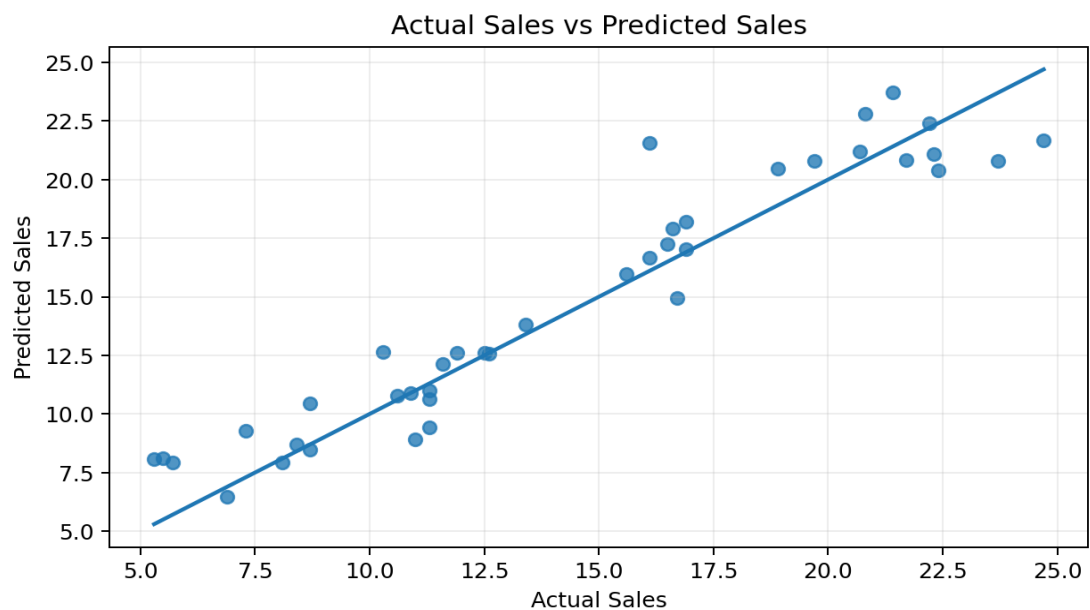
6. PLANNED ADVERTISING BUDGET

Advertising Medium	Planned Budget
TV	150
Radio	20
Newspaper	30

Predicted Sales = 15.04 units

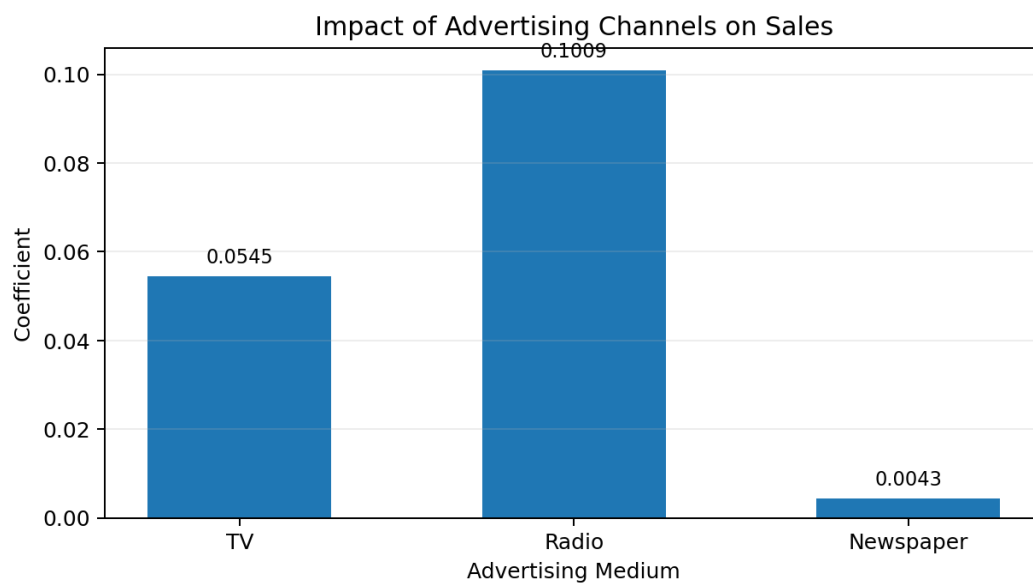
This is the model's estimated sales for the given advertising budget.

7. ACTUAL VS PREDICTED SALES



The points are generally close to the diagonal reference line, showing that the predicted values are reasonably close to the actual sales values in the test set.

8. ADVERTISING CHANNEL IMPACT



Based on the regression coefficients, **Radio** has the strongest impact (0.1009), while **Newspaper** has the least impact (0.0043). TV also has a positive contribution (0.0545).

Findings, Recommendations & Conclusion

9. KEY FINDINGS

Question	Finding
Strongest impact	Radio has the highest raw regression coefficient.
Least impact	Newspaper has the smallest coefficient.
Model quality	$R^2 = 0.9059$; the model explains most of the variation in unseen sales.
Planned budget prediction	TV 150 + Radio 20 + Newspaper 30 → 15.04 predicted sales.

10. BUSINESS RECOMMENDATION

Focus more attention on Radio advertising.

The model gives Radio the strongest regression coefficient. The company can optimize its Radio budget and monitor whether the change is followed by improved sales.

11. TECHNICAL IMPROVEMENT

Test stronger models and improve validation.

Random Forest or Gradient Boosting can be compared with Linear Regression. Cross-validation and hyperparameter tuning can also be used to select a more accurate model.

12. CONCLUSION

The analysis shows a positive relationship between the advertising channels and Sales. A Multiple Linear Regression model was trained using historical data and achieved a test R^2 of 0.906. Radio has the strongest coefficient, while Newspaper has the least. For the planned budget of TV = 150, Radio = 20, and Newspaper = 30, the predicted sales are approximately 15.04 units. The model provides a useful baseline, while other machine learning models and better validation can be explored for further improvement.

Report prepared from the submitted Advertising Sales Prediction Jupyter Notebook and dataset.